TEXORA · LECTURE SHEET

Process Control in Ring Frame

The ring frame is where count, strength and ends down are finally decided — and where most of the mill's spindles, labour and cost sit. This sheet covers count control and its draft correction, count-strength product (CSP), ends-down measurement and Pareto analysis, and the traveller and climate controls behind them.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 8 · SPECIALIZATION — YARN MANUFACTURING

1. Count control

Count is tested on leas from many cops (Topic 6 of YM-3207 gives the lea method). The reported values are mean count, count CV%, and the deviation from nominal. Control practice: sample across the whole frame and across time, because a frame can be on-count at the headstock end and off at the tail through roving supply differences.

WORKED EXAMPLE — COUNT DEVIATION, CV AND DRAFT CORRECTION

Nominal count = Ne 30.0. Twenty lea tests give mean Ne 29.2 with SD 0.5.

Deviation = (30.0 − 29.2)/30.0 × 100 = **2.67%** — the yarn is coarser than nominal (lower Ne = heavier yarn).

Count CV = 0.5/29.2 × 100 = **1.71%** — inside the usual ≲ 2% expectation for a controlled frame.

Correction: count is inversely proportional to total draft. To move the mean from 29.2 to 30.0, raise draft by the ratio 30.0/29.2 = 1.0274, i.e. **+2.7%** on the draft change wheel, then retest.

Before correcting, verify the roving: if roving is 2.7% heavy (Topic 7), the correct action is at the simplex, not the ring frame. Count control is a chain discipline, not a per-machine reflex.

2. Strength: CSP

Yarn strength is compared between counts through the count–strength product of a lea test:

$$\text{CSP} = \text{lea strength (lbf)} \times \text{count (Ne)}$$

WORKED EXAMPLE — CSP

Lea strength = 78 lbf at the tested count of 29.2.

CSP = 78 × 29.2 = **2278**.

For medium-staple cotton at Ne 30, CSP values around 2200–2400 indicate a sound mixing spun well; a fall in CSP with count on target redirects investigation to the mixing (fibre strength, Topic 2) or to twist (below), not to the draft.

Twist control: actual twist is verified by untwist–retwist testing; twist below the TM target (3.9 in the YM-3207 example) costs strength and raises ends down, while excess twist costs production (delivery falls as TPI rises) and makes harsh yarn. Twist CV across spindles exposes worn spindle drives and traveller problems.

3. Ends down — measurement and Pareto analysis

Ends down rate is the ring frame's vital sign, quoted per 100 spindle-hours:

$$\text{Ends down / 100 spindle-hours} = \text{breaks counted} / (\text{spindles watched} \times \text{hours}) \times 100$$

WORKED EXAMPLE — ENDS DOWN RATE

A patrol counts 120 breaks on a 1000-spindle frame over 12 hours.
Rate = 120 / (1000 × 12) × 100 = **1.0 per 100 spindle-hours** — a healthy level for Ne 30 cotton.
The rate alone does not fix anything; the **causes** recorded with each break do. A shift audit of 250 breaks classified them as follows:

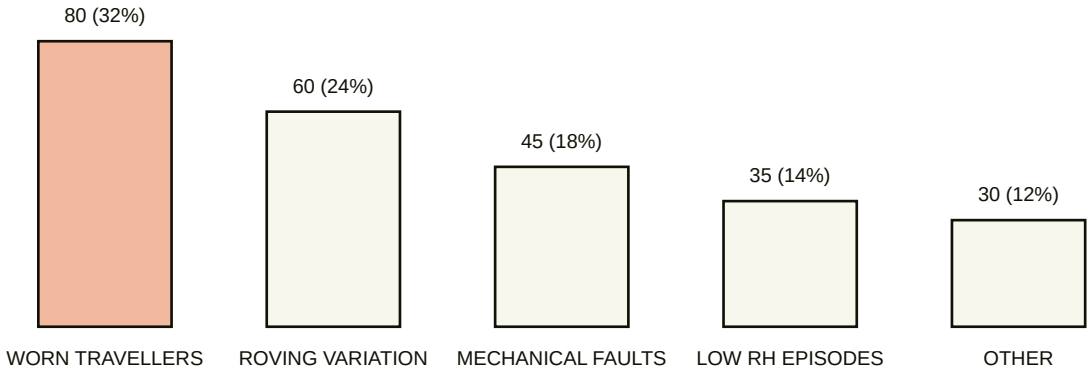


Figure 1 — Ends-down Pareto chart (bar heights to scale). Process control works the tallest bar first: traveller changing discipline alone addresses a third of the breaks.

4. The working controls behind the numbers

CONTROL	STANDARD PRACTICE	SYMPTOM WHEN NEGLECTED

Traveller changing	Changed on a running-hours schedule, correct weight/profile for count and speed	Ends down climb; yarn hairiness and strength fall
Ring and spindle condition	Rings centred, spindles true, no bent skewers	Localised breaks at repeated spindle positions
Roller cots and aprons	Buffed/replaced on schedule; shore hardness checked	Drafting faults, thick/thin places, CSP fall
Pneumafil / suction	Ducts clear, suction verified at frame end	Roller lapping, fly accumulation, breaks
Climate	RH held at department standard (Topic 13)	Breaks spike in dry episodes — visible in the Pareto above
Roving supply	Bobbin identity and age controlled; no mixed lots on one frame	Count drift along the frame, unexplainable breaks

Glossary

CSP — count–strength product: lea strength (lbf) × Ne, the standard strength comparator. **Ends down** — yarn breaks at the ring frame, quoted per 100 spindle-hours. **Pareto analysis** — ranking causes by frequency so the largest is attacked first. **Traveller** — the C-shaped clip riding the ring flange; inserts twist and controls balloon tension, and is a consumable. **Balloon** — the rotating yarn loop between guide eye and traveller. **Pneumafil** — the suction system clearing broken ends and fly at the drafting zone. **Lapping** — fibre winding onto a roller after an end break instead of passing forward.

Key points

• Count example: mean 29.2 vs nominal 30.0 = 2.67% coarse, CV 1.71%; correction +2.7% draft — after verifying the roving.
• CSP = lea strength × count = 2278 in the example; it separates strength problems from count problems.
• Ends down = breaks/(spindles × hours) × 100; 1.0 per 100 spindle-hours is healthy for Ne 30 cotton.
• Pareto the causes: travellers (32%) and roving variation (24%) own 56% of breaks in the audit.
• Traveller life, cot condition, pneumafil and RH are the standing controls behind count and breaks.

Practice

1. Lea strength is 92 lbf at a tested count of Ne 36.2. Calculate the CSP.

2. A 1200-spindle frame records 216 breaks in 9 hours. Calculate the ends down per 100 spindle-hours and judge it against the level in this sheet.
3. Count tests along a frame read on-target at the headstock but progressively coarser toward the tail. Where do you look first, and why not at the draft wheel?

Answer key

1. CSP = 92 × 36.2 = **3330**.
2. Rate = 216/(1200 × 9) × 100 = **2.0 per 100 spindle-hours** — double the healthy example level; start a cause-recorded patrol and Pareto the results before changing settings.
3. At the roving supply along the frame: bobbins at the tail are likely from a different (heavier) roving lot or an older simplex doff. The draft wheel is common to the whole frame — it cannot produce a gradient along it — so a positional pattern proves a supply or local mechanical cause, not a setting error.

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